

Fitting absolutely monotonic real functions in bounded intervals

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We discuss the problem of fitting absolutely monotonic functions to noisy random samples in bounded intervals with both symmetric and asymmetric noise and target loss. This specific shape-constrained estimation problem presents a unique set of constraints that challenge the capabilities of most existing approaches. We explore the applicability and limitations of various kinds of shape-constrained estimation, including exponential sum fitting, constrained Bernstein polynomials, constrained splines, and variations of traditional Vandermonde estimation. A novel approach, Greedy Lowballing Absolutely Monotonic Polynomial Fitting (GLAMPF) is also introduced. This method finds monomial terms to iteratively approximate the sampled data from below (lowballing), guaranteeing absolute monotonicity by construction, demonstrating robustness to a variety of function shapes, and exploiting underestimation bias for algorithmic structure. However, it lacks general optimality and occasionally struggles to align with the various regimes of slow and fast growth. A few of the most promising methods were implemented and compared on generated samples from a wide variety of absolutely monotonic and convex functions. While convex splines displayed the best results in cases where low degree polynomials were sufficient, GLAMPF was the most robust approach with the strongest guarantees and produced acceptable results in every test case.

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1 Introduction

Shape constraints are used when fitting real functions to data points in order to enforce certain growth, geometric, or analytical properties on the estimated functions, that are assumed to hold for the underlying true function and/or represent useful properties of the goal solution to the fitting problem. The most typical shape constrained families of functions include *increasing* and *convex* functions. Shape constraints can have multiple related characterizations. For example, smooth increasing and convex functions can be characterized by positive first and second derivatives, respectively, or by geometric constraints that do not require derivatives. Convex smooth functions,

and in particular fitting convex smooth functions to data, have been studied extensively [Groeneboom et al., 2001, Lu, 2015, Lebair and Shen, 2015, Nagahara and Martin, 2013, Zhang, 2004, He and Shi, 1998, Aguilera and Morin, 2008, Hannah and Dunson, 2013, Holmström and Petersson, 2002, Lachand-Robert and Oudet, 2005, Kopotun et al., 2011, Wang and Ghosh, 2012, Ghosal et al., 2023] due to the multitude of applications in economics, statistics, machine learning, etc.

A real function is *absolutely monotonic* if all of its derivatives are non-negative. Absolutely monotonic functions are therefore increasing and convex, but not all increasing convex functions are absolutely monotonic. Absolute monotonicity is significantly restrictive, and it may appear in applications with natural strong growth patterns. For example, a function that represents the reward or performance associated with an accumulative score in a videogame level would not only need to be increasing, but may also need to significantly reward scores approaching perfect execution in a way that represents the fast and increasingly accelerating difficulty of approaching perfection closer and closer, and may be fit from data points of player performances labelled with player skill levels or proxy measurements relating to player skill. Absolutely monotonic shape-constrained estimation from fixed samples has not been studied in the same level of detail that convex estimation has, primarily due to the rigidity of absolutely monotonic constraints and that convexity is deemed sufficient for most applications. The majority of methods that ensure absolute monotonicity of the resulting functions do so *by construction*, and usually as a side effect of tighter restrictions. For example, fitting exponential sums [Holmström and Petersson, 2002, Keller and Plonka, 2021, Chunaev and Safiullin, 2020, Evans et al., 1980, Smith et al., 1976, Kammler, 1976] or Vandermonde polynomial fitting [Björck and Pereyra, 1970] restricted to positive monomial coefficients. These also sometimes presume we can select the points to sample on.

The goal of this paper is to discuss our investigation into potential approaches to fit absolutely monotonic functions to data, their strengths, weaknesses, and some simulated results. This discussion highlights the tension between less constrained methods that do not guarantee absolute monotonicity, and methods that do guarantee absolute monotonicity but are not particularly well suited for fitting from random samples. We also introduce one new method, Greedy Lowballing Absolutely Monotonic Fitting (GLAMPF), that builds on the structure of the problem while avoiding assumptions required by other methods to produce results that are frequently not optimal, but consistently reliable, as opposed to most other methods that have catastrophic failure modes.

§2 provides a more formal description of the problem. §3 discusses relevant mathematical connections and facts. §4 explains the attempted approaches, with their strengths and weaknesses, including the implementation of those that were implemented. §5 presents the simulation and evaluation set up for the comparison of the implemented approaches. §6 discusses the results of the evaluation and general discussion on how different methods compare. §7 presents the conclusions from this paper.

2 Problem statement

Given a single variable **real valued** set of datapoints in a **bounded interval** (we assume $[0, 1]$ without loss of generality), we consider the problem of finding **absolutely monotonic** functions that best approximate the data. We will, however, compare with some methods that give weaker guarantees than absolute monotonicity, due to their prominence and to understand their relative advantages and disadvantages.

We presume noise in the data, and we evaluate two different situations with respect to noise and related loss function, paying attention to the flexibility of the discussed methods with respect to unconventional optimization targets.

- First, we consider **normally distributed noise** proportional to the function to model symmetric noise from independent sources on the data, and use a traditional **square of residuals** loss function to fit this data.

$$y_i|x_i \sim f(x_i) + \mathcal{N}(0, \sigma) \quad (1)$$

$$l_1(r) = r^2 \quad (2)$$

- Second, we consider **positive exponentially distributed noise**. This can represent situations, such as the aforementioned videogame score reward case, where the target function by definition relates to best possible performances, and therefore underperforming data points would lead to positive error in the absolutely monotonic function. If the x axis is score obtained and y axis is player skill, a high-level player underperforming would have a higher player skill than their score suggests, and the target function corresponds to the best possible performances of players of a certain skill. We introduce an **asymmetric exponential penalty** to the loss function in order to fit this data.

$$y_i|x_i \sim f(x_i) + \text{Exp}(\lambda) \quad (3)$$

$$l_2(r) = |r| + e^{-tr} \quad (4)$$

for $t > 0$.

In our discussion of potential approaches, we also consider their *parametric*, *semi-parametric*, or *non-parametric* nature. Parametric or semi-parametric methods may be more adequate for applications where a small, bounded number of parameters to express the resulting function is required.

The summarized problem statement is as follows, and leaves some room for variability to consider the different approaches' advantages and disadvantages.

Problem 2.1. *Given a set of real valued datapoints $\{x_i, y_i\}$, where $x_i \in [0, 1]$, find a function $f \in C^\infty([0, 1])$ that:*

- *is absolutely monotonic in $[0, 1]$.*
- *minimizes the average loss $l(y_i - f(x_i))$.*
- *can ideally be represented (semi-)parametrically with a bounded, low number of parameters.*

3 Exponential mixtures and other characterizations of absolutely monotonic functions

Before discussing potential approaches for fitting absolutely monotonic functions, it is relevant to discuss the mathematics behind a couple of characterizations of absolutely monotonic functions, and particularly the significant connection with exponential functions and mappings.

3.1 Bernstein's characterization of absolutely monotonic functions in $(-\infty, 0]$

Exponential functions with positive exponents are always absolutely monotonic. One deeply related family of functions are *completely monotonic functions*, which are functions with alternating sign derivatives. It is easy to see that $f(x)$ is absolutely monotonic in (a, b) if and only if $f(-x)$ is completely monotonic in $(-b, -a)$. Moreover, Bernstein's theorem characterizes all completely monotonic functions in $[0, +\infty)$, and consequently also all absolutely monotonic functions in $(-\infty, 0]$, as sums of exponentials:

Theorem 3.1 ([Bernstein, 1929]). *Let $f \in \mathcal{C}^\infty([0, +\infty))$. Then, f is completely monotonic if and only if it is the Laplace transform of a non-negative finite Borel measure μ . Explicitly:*

$$f(x) = \int_0^\infty e^{-tx} d\mu(t) \quad (5)$$

or, rephrasing it for absolutely monotonic functions:

Theorem 3.2 ([Bernstein, 1929]). *Let $f \in \mathcal{C}^\infty((-\infty, 0])$. Then, f is absolutely monotonic if and only if there is a non-negative finite Borel measure μ such that:*

$$f(x) = \int_0^\infty e^{tx} d\mu(t) \quad (6)$$

In a few words, any absolutely monotonic function in $(-\infty, 0]$ can be uniquely characterised as a mixture of exponentials with positive exponents. This also forces f to be analytic due to its analytic representation. The mathematical theory of completely monotonic, absolutely monotonic, and related families of functions is much broader and deeper than what we discuss or use in this paper [Koumandos, 2014].

3.2 Logarithmic convexity

Another important family of shape-constrained functions are *logarithmically convex* functions, which are functions whose logarithm is convex. Logarithmically convex functions must be convex, and they also appear in a multitude of applications. Exponentials of the form e^{tx} are one of the simplest subfamilies of logarithmically convex functions. Moreover, logarithmically convex functions (like increasing, convex, and absolutely monotonic functions) form a cone and are therefore closed under sum/mixture. It follows from this and Bernstein's theorem that an absolutely monotonic function in $(-\infty, 0]$ must be logarithmically convex.

Lemma 3.1. *Let $f \in \mathcal{C}^\infty((-\infty, 0])$ be absolutely monotonic. Then, f must be logarithmically convex.*

Proof. By Bernstein's theorem, $f(x) = \int_0^\infty e^{tx} d\mu(t)$ for some measure μ , but each e^{tx} is logarithmically convex, and a mixture of logarithmically convex functions must be logarithmically convex [Boyd and Vandenberghe, 2004]. $\square$

That is, in the half-line $(-\infty, 0]$, absolute monotonicity is strictly stronger than logarithmic convexity. This, however, does not hold in bounded intervals such as $[0, 1]$. Both absolutely monotonic and logarithmically convex families become significantly less restrictive in bounded intervals, in different ways. For example, the following can be shown.

Lemma 3.2. *Let $f \in \mathcal{C}^\infty([0, 1])$ be strictly convex (in the sense that $f''(x) > 0$ for all $x \in [0, 1]$). Then, there is a constant C such that $f + C$ is logarithmically convex in $[0, 1]$.*

Proof. The second derivative of the logarithm of f is $(\log f)'' = \frac{ff'' - (f')^2}{f^2}$. In order for $(\log f)'' > 0$, with $f'' > 0$, it is necessary and sufficient that the numerator of this fraction is greater than 0. That is:

$$ff'' > (f')^2 \quad (7)$$

All derivatives of $f + C$ are equal to those of f , and therefore for the function $f + C$ to be logarithmically convex, it is necessary and sufficient that

$$(f + C)f'' > (f')^2 \quad (8)$$

but then we can solve for C :

$$C > \frac{(f'(x))^2}{f''(x)} - f(x) \quad (9)$$

for all $x \in [0, 1]$. But since $[0, 1]$ is a bounded interval, then there must be a supremum to the right hand side of equation 9. And thus, we may choose any constant C greater than the supremum, to ensure that $f + C$ is logarithmically convex. $\square$

The significance of this is that logarithmic convexity in bounded intervals is less of a shape constraint and more of a vertical requirement on top of traditional convexity, and yet, exponential sums continue to be canonical representatives of both logarithmically convex and absolutely monotonic families.

3.3 Absolutely monotonic functions in bounded intervals

Many absolutely monotonic functions in $[0, 1]$ cannot be extended analytically to $(-\infty, 0]$, and therefore do not admit an exponential sum representation. A simple and important example of this is monomials $f(x) = x^n$. All

of these are absolutely monotonic (and, in fact, analytic) in $[0, 1]$, but cannot be extended to an absolutely monotonic function in $(-\infty, 0]$, and cannot be represented as sums of exponentials (they are also not logarithmically convex in $[0, 1]$).

However, there is an important connection between monomial representations in $(0, 1]$ and exponential representations in $(-\infty, 0]$. Consider the diffeomorphism $x = e^y$ that identifies functions $f(x) \in \mathcal{C}^\infty((0, 1])$ with functions $g(y) \in \mathcal{C}^\infty((-\infty, 0])$. First, we note that absolute monotonicity of f is strictly stronger than absolute monotonicity of g .

Lemma 3.3. *Let $f(x) \in \mathcal{C}^\infty((0, 1])$ be an absolutely monotonic function and consider $g(y) = f(e^y) \in \mathcal{C}^\infty((-\infty, 0])$.*

Then:

1. *g is absolutely monotonic.*
2. *The Bernstein representation $g(y) = \int_0^\infty e^{ty} d\mu(t)$, must be such that μ is supported only on integer t , that is, $\mu = \sum_{t=1}^\infty a_t \delta_t$, and thus*

$$\begin{aligned} g(y) &= \sum_{t=0}^\infty a_t e^{ty} \\ f(x) &= \sum_{t=0}^\infty a_t x^t \end{aligned} \tag{10}$$

with $a_t \geq 0$.

Proof. Proving g is absolutely monotonic is a straightforward application of the chain rule:

$$\begin{aligned} g'(y) &= (f(e^y))' = f'(e^y) \cdot e^y \\ g''(y) &= f''(e^y) \cdot e^y \cdot e^y + f'(e^y) \cdot e^y \\ g'''(y) &= f'''(e^y) \cdot e^y \cdot e^y \cdot e^y + 3f''(e^y) \cdot e^y \cdot e^y + f'(e^y) \cdot e^y \\ &\dots \end{aligned} \tag{11}$$

where we can see that subsequent derivatives only include sums and products of derivatives of f evaluated in $e^y \in (0, 1]$ (positive because f is absolutely monotonic) and e^y terms (always positive), and therefore are all positive.

From this, we can apply Bernstein's theorem to present $g(y) = \int_0^\infty e^{ty} d\mu(t)$ for non-negative Borel measure μ , and apply the change of variables $x = e^y$, then $f(x) = f(e^y) = g(y) = \int_0^\infty e^{ty} d\mu(t) = \int_0^\infty x^t d\mu(t)$. Consider the derivatives of f under this representation:

$$f^{(n)}(x) = \int_0^\infty t \cdot (t-1) \cdot \dots \cdot (t-n+1) x^{t-n} d\mu(t) \tag{12}$$

which we know to be positive for all n . For integer t , these subsequent multiplications are all positive until a 0 term appears (after t derivations), cancelling out the term in all subsequent derivatives. This is what we expect with integer degree polynomials.

For non-integer t , the term $t \cdot (t-1) \cdot \dots \cdot (t-n+1)$ changes sign as n increases and grows factorially in magnitude. If μ assigns positive density to any non-integer t , then for sufficiently large n , the integral contains contributions of alternating sign and unbounded magnitude that cannot be canceled by a fixed non-negative measure. This contradicts the requirement that all derivatives remain non-negative. Therefore, $\mu = \sum_{t=1}^{\infty} a_t \delta_t$.

An alternative, perhaps more formal proof can be provided by relying on the fact that absolutely monotonic functions must be analytic. From this, we know that $f(x) = \sum_{t=0}^{\infty} a_t x^t$. We can pull this back to the exponential representation and match it with Bernstein's representation.

$$g(y) = \int_0^{\infty} e^{ty} d\mu(t) = \sum_{t=0}^{\infty} a_t e^{ty} \quad (13)$$

and use uniqueness of Laplace transforms to show that it must be $\mu = \sum_{t=1}^{\infty} a_t \delta_t$. $\square$

This result works in both directions, though, and we can show that discrete integer sums is exactly the condition required to map absolutely monotonic functions in $(-\infty, 0]$ to absolutely monotonic functions in $(0, 1]$.

Lemma 3.4. *Let $g(y) \in \mathcal{C}^{\infty}((-\infty, 0])$ be an absolutely monotonic function that only has integer terms in its Bernstein representation. That is, $g(y) = \sum_{t=0}^{\infty} a_t e^{ty}$.*

Then, $f(x) = g(\log x) \in \mathcal{C}^{\infty}((0, 1])$ is absolutely monotonic.

Proof. Simply write f as a discrete sum. $f(x) = g(\log x) = \sum_{t=0}^{\infty} a_t e^{t \log x} = \sum_{t=0}^{\infty} a_t x^t$. We know that for integer $t > 0$, each x^t is absolutely monotonic, and thus their discrete positive sum is also absolutely monotonic. $\square$

In summary

Theorem 3.3. *There is a one-to-one correspondence between absolutely monotonic functions in $(0, 1]$ and their Taylor series around 0, $f(x) = \sum_{t=0}^{\infty} a_t x^t$; and absolutely monotonic functions in $(-\infty, 0]$ with discrete Bernstein measures supported only on the integers, $g(y) = \sum_{t=0}^{\infty} a_t e^{ty}$. This correspondence is given by the change of variables $x = e^y$.*

For our problem, this means that fitting absolutely monotonic functions in $(0, 1]$ by fitting sums of monomials can be translated to and from fitting exponential sums in $(-\infty, 0]$, constrained to integer exponents.

4 Potential approaches

4.1 Positive sums of positive exponentials

The methods discussed in this section find solutions of the form $f(x) = \sum a_t e^{tx}$ (possibly with affine transformations). As discussed in §3, these sums are absolutely monotonic by construction, and are representative approximations of absolutely monotonic functions in $(-\infty, 0]$.

Prony’s method relies on sampling at prescribed intervals to exploit the structure of complex exponential sums to reduce the fitting problem to finding roots of an associated polynomial [Holmström and Petersson, 2002, Keller and Plonka, 2021, Chunaev and Safiullin, 2020]. This is an effective method for decomposing known functions that can be sampled at prescribed intervals into sums of complex exponentials, but is not suitable for fitting a function to random data that does not adjust to this structure (complex exponential sums are not absolutely monotonic) and contains noise.

Non-linear least squares optimization with various suitable setups can be used to approximate a set of parameters (both coefficients a_t and exponents t) for an exponential sum that minimizes loss on the datapoints [Holmström and Petersson, 2002, Evans et al., 1980]. This presents challenges such as convergence to local optima (particularly for unusual loss functions), high reliance on the parameterization choice, complex and unpredictable effects of noise in the data, the requirement for the loss function to be smooth, and poor computational scalability to higher number of parameters. It is nonetheless a feasible option for datapoints that can be approximated using exponential sums.

A number of other approaches and variations on the aforementioned approaches for fitting exponential sums exist. All of these share some of the challenges discussed, especially the need for specific sampling, sensitivity to noise, convergence to local optima, and computational issues. For example, [Smith et al., 1976] uses a spectral method to fit exponential sums, but the approach is outdated, computationally limited, shows mixed results, and most importantly, requires admitting negative coefficients, losing absolute

monotonicity.

4.2 Constrained piecewise linear interpolation

While exponential sums ensure absolute monotonicity *by construction*, a very different approach uses *explicit constraints* to enforce monotonicity. An extreme version of this is to build piecewise linear interpolations that minimize loss with respect to the data points, under, for example, convexity constraints. Convexity can be expressed as optimization constraints by indicating that the second order discrete differences must all be positive:

$$\frac{y_{i+2} - y_{i+1}}{x_{i+2} - x_{i+1}} - \frac{y_{i+1} - y_i}{x_{i+1} - x_i} \geq 0 \quad (14)$$

Piecewise linear interpolation is highly flexible, and therefore can produce low loss fits, and is achievable efficiently with a wide variety of optimization algorithms, but it has a number of important issues. First, higher-order monotonicity constraints involve more and more complex optimization constraints, with strict absolute monotonicity not being possible to express (but rather up to an arbitrary degree of discrete differences). The more higher-order constraints are used, the less the computational attractiveness of the optimization algorithm can be exploited. Moreover, piecewise linear interpolation is only continuous and not differentiable. Finally, this method is inherently nonparametric, with the result being dependent on the number of interpolating points used, meaning that using a small number of parameters will lead to worse fits.

We implemented this method (with basic convexity constraints) for comparison purposes even if it does not satisfy important conditions of our problem.

4.3 Polynomials

In §3 we discussed that absolutely monotonic functions in $[0, 1]$ must be positive monomial sums. In this section we discuss methods that find solutions of the form $f(x) = \sum a_t x^t$.

Some important questions remain, and different polynomial fitting methods will be discussed in this section to address these questions. The first is how to obtain good approximation coefficients a_t from the sample. The second is how the fitting method responds to noisy data. Finally, there is

the question of the number of coefficients used in the expression, and how this relates to a good approximation.

Spectral methods are impractical for this problem. Taylor series are sensitive to small variations and overall lack a stable structure that preserves coefficients and/or degree dominance under approximation circumstances, an issue that grows the further away from 0 that we evaluate the series. From a practical point of view, it is challenging, especially from sampled noisy data, to reliably reconstruct an exact and/or uniquely defined Taylor series, or to isolate the largest coefficients in the Taylor series. Taylor polynomials with different coefficient distributions can be similar pointwise, meaning that a preferred coefficient or degree from data points alone tends to be unstable under noise.

4.3.1 Bernstein polynomials

In certain ways the closest approach to spectral methods, Bernstein polynomials offer a promising approach to our problem. Bernstein polynomials are of the form:

$$b_{k,N}(x) = \binom{N}{k} x^k (1-x)^{N-k} \quad (15)$$

While Bernstein polynomials are not absolutely monotonic, work like [Wang and Ghosh, 2012, Ghosal et al., 2023] present explicit methods to perform shape-constrained regression using Bernstein polynomials from sampled data, with strong approximation and performance guarantees. Specifically, Bernstein basis polynomials are a basis for the vector space of polynomials of a certain degree with important interpolation properties. [Wang and Ghosh, 2012] showed that polynomial regression of a certain degree with shape constraints can be solved by using an adequate matrix representing the shape constraints (monotonicity, convexity, etc.) on the discrete differences applied to the Bernstein basis. With a typical squared residual objective this can then be solved using linear least squares optimization.

However, this has a number of challenges for application to our problem. First, no matrix expression for absolute monotonicity can be found in the literature. Nonetheless we believe it would be possible to express absolute monotonicity up to the maximum derivative that matters with adequate matrices, but, similarly to non-linear least squares for exponential sums (§4.1), these would increase the size of the optimization problem substantially, as

it involves explicit constraints rather than absolute monotonicity by construction. More importantly, the method is constrained from inception to a maximum degree of polynomials. In absolutely monotonic fitting, especially when the underlying function has extreme growth patterns, we may need very high degree polynomials to adequately represent it, even for small samples. The Bernstein polynomial linear method requires a pre-determined maximum degree, and the complexity of the optimization problem increases with the degree (so large degrees are not feasible). Finally, this is inherently a nonparametric method that requires a quite large set of coefficients dependent on the degree. The basis is not set up in a way that allows removing some of the elements while preserving a significant proportion of fitting capabilities.

Another approach using Bernstein polynomials would be feasible if the function could be sampled at specific intervals, following Bernstein’s original proof of the Weierstrass approximation theorem [Bernstein, 1912], but as discussed this is also infeasible for our problem due to the random sample.

4.3.2 Constrained least squares Vandermonde

A simple yet important to compare to approach uses the Vandermonde matrix over the dataset up to a maximum degree to present a linear optimization problem to find a polynomial estimation.

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad (16)$$

However, in order to enforce absolute monotonicity, non-negativity constraints need to be enforced on the matrix. This makes linear least squares unsuitable to solve the problem due to the severe restriction of the parameter space. Furthermore, the use of a custom loss function in our problem forces the use of non-linear least squares optimization. Moreover, high degree polynomials are required to represent absolutely monotonic functions well enough, increasing the complexity of the optimization problem. Perhaps much more importantly, Vandermonde interpolation is numerically unstable and badly represented for high degrees, with high sensitivity to noise or numeric imprecision.

cisions during optimization.

We implemented this method for comparison purposes even if it does not satisfy important conditions of our problem.

4.3.3 Convex splines

A popular family of shape-constrained methods discussed extensively in the literature use spline interpolation [Lu, 2015, Lebar and Shen, 2015, Nagahara and Martin, 2013, Zhang, 2004, He and Shi, 1998]. Similar to Bernstein polynomials, these methods use explicit constraints on an optimization problem to ensure monotonicity constraints. They have similar issues to Bernstein polynomials for application to our problem. Absolute monotonicity constraints are not discussed in the literature and would be even harder to implement for spline interpolation, if possible at all. Degree constraints are even harsher than Bernstein polynomial methods, with a degree of 3 used frequently in the literature, which is considered insufficient to represent functions with significantly more aggressive growth patterns. They are also nonparametric with the number of parameters being highly correlated with the quality of the approximation.

This method is more promising in terms of approximation capability and performance than other methods and relatively easy to implement, so we implemented it with degree 3 and convexity constraints for comparison, even if it does not satisfy all of the conditions of our problem.

4.3.4 Greedy lowballing monomial sums

We are not aware of literature on the use of greedy algorithms for polynomial fitting in similar problems. To a large degree, this makes sense: the general polynomial fitting problem is unstable and must be solved holistically for satisfactory solutions, lacking any significant stable structure that can be exploited to iteratively build approximations that consistently improve. Moreover, without further constraints, global solvers, including some of the ones discussed so far, work efficiently and effectively, so there is no reason for slower methods with fewer guarantees.

However, in the particular context of the problem presented in this paper, we argue that a greedy approach is attractive. From the point of view of the (semi-)parametric goal, producing a polynomial one monomial at a time slowly approaching the target allows us to select a limited number of

monomials to include in the final solution, reducing the quality of the approximation in a measured way in order to obtain a simpler solution expression. Perhaps more importantly, the absolutely monotonic constraints lend themselves to an iterative *lowballing* approach, where we slowly add terms, always underestimating the solution by smaller and smaller amounts at each step, approaching the solution from underneath. Unlike general polynomial fitting, the positivity and underestimation constraints create a structure where partial sums remain valid approximations.

We thus introduce the Greedy Lowballing Absolutely Monotonic Polynomial Fitting (GLAMPF) algorithm. This algorithm is highly specialised to this problem and has significant disadvantages, especially if used for more general tasks, compared to the other methods discussed in this section, but is particularly attractive to our problem in particular. We can summarize its key features as follows:

- The algorithm has two sets of parameters: *degrees* and *coefficients*, which are optimized simultaneously (one pair at a time). Their number is not necessarily predetermined but can be limited with measured effects, making the method semi-parametric.
- At each iteration, we add a new monomial with a certain degree and coefficient, of the form $c_j x^{d_j}$, that minimizes the loss with respect to the remainder of the target. After each iteration, we subtract the current monomial from the target value at each point.
- Once an optimal degree and coefficient has been found for a given iteration, we reduce its contribution by a factor $\gamma < 1$ before subtracting it, to ensure the *lowballing*. Together with the asymmetric loss, this ensures that the target is always underestimated, but built towards in steps that best represent the structure of the remainder. That is, we do $y_i^{j+1} = y_i - \gamma c_j x_i^{d_j}$. This lowballing aspect means that the same degree may need to be added multiple times, but gives space for changing the preferred degree in subsequent iterations without overcommitting to a single one too early, ensuring a more balanced mixture that approximates the function better across the domain.
- γ is increased at each iteration, approaching 1 over time, to slowly converge to a full solution.
- We accumulate the coefficients and degrees, and once a certain threshold of approximation or a threshold for small coefficients are surpassed,

the algorithm stops. We filter excessively small resulting coefficients and/or limit to a certain number of highest coefficients for an efficient parametric final expression. We note that $0^d = 0$ and $1^d = 1$ for all d , which means that the coefficients of different degrees are comparable in scale when fitting in the $[0, 1]$ interval: different degrees merely represent different shapes (higher degree corresponding to more curved and concentrated growth towards 1).

Among important disadvantages of this approach compared to other discussed ones, we include computational inefficiency due to the greedy nested optimization pattern and dependence on hyperparameter choice and data particularities in the overall result. However, unlike all the other discussed methods, this method simultaneously ensures absolute monotonicity by construction, is not bound to a low predetermined maximum degree, behaves particularly well with asymmetric loss due to its inherent asymmetric nature, and produces semi-parametric outputs that can easily be limited to a small number of parameters with measured effects.

We implemented this method and consider it the one to have displayed the most robust, if not optimal, results.

4.4 Summary

The following table qualitatively summarizes our discussion on the considered approaches and explain our choices.

GLAMPF has good representation of the target function family and absolute monotonicity guarantees. The only other methods with good qualities in these columns are Prony methods and precisely sampled Bernstein polynomials, which are completely infeasible due to fixed sampling, nonparametric nature, and no space for an asymmetric loss function; as well as non-linear least squares exponential sums, which have issues with noise sensitivity and asymmetric loss functions. In exchange, our method does not guarantee optimality, is sensitive to hyperparameters, and is computationally inefficient.

Non-linear least squares exponential sums and Vandermonde interpolation with positivity constraints are the only two other methods with acceptable combinations of properties for our problem, and both have significant disadvantages with respect to GLAMPF. To the best of our knowledge, no existing method simultaneously satisfies absolute monotonicity by construction, robustness to noisy non-uniform samples, compatibility with asymmetric loss, and a low-dimensional parametric representation.

Table 1: Summary of potential approaches to fitting absolutely monotonic functions and their properties

Method	Representation	Monotonicity	Optimality	Hyperparams	Noise	Loss	Sampling	Parametric	Performance
Prony methods §4.1	● Complete with appropriate change of variables and constraints	● Absolutely monotonic if constrained to positive real exponents	✓ Yes	✓ Irrelevant	✗ Highly sensitive	✗ Fixed target	✗ Fixed	✓ Parametric	✓ Fast
Non-linear least squares exponential sums §4.1	● Complete with appropriate change of variables and constraints	✓ Absolutely monotonic, by construction	⚠ Local optima	⚠ Sensitive	⚠ Sensitive	● Flexible, but affected	✓ Any	✓ Parametric	⚠ Problematic
Constrained piecewise linear interpolation §4.2	✗ Not differentiable	⚠ Convex	✓ Yes	✓ Irrelevant	● Robust	✓ Flexible	✓ Any	✗ Nonparametric	✓ Fast
Shape-constrained Bernstein polynomials §4.3.1	⚠ Limited degree	⚠ Explicit constraints, absolutely monotonic difficult	✓ Yes	⚠ Sensitive	⚠ Sensitive	● Flexible, but affected	✓ Any	✗ Nonparametric	⚠ Affected by degree and constraints
Precisely sampled Bernstein polynomials §4.3.1	● Selected degree	● Inherited from source function	✓ Yes	✓ Irrelevant	✗ Highly sensitive	✗ Fixed target	✗ Fixed	✗ Nonparametric	● Moderate
Constrained Vandermonde §4.3.2 optimization	⚠ Limited degree	✓ Absolutely monotonic, by construction	⚠ Local optima	⚠ Sensitive	⚠ Sensitive	● Flexible, but affected	✓ Any	● Semi-parametric	⚠ Problematic, due to constraints
Convex splines §4.3.3	⚠ Very limited degree	⚠ Convex	✓ Yes	⚠ Sensitive	● Robust	● Flexible, but affected	✓ Any	✗ Nonparametric	⚠ Affected by degree and constraints
Greedy low-balling monomial sums §4.3.4	● Highly flexible, constrained by number of parameters	✓ Absolutely monotonic, by construction	⚠ No	⚠ Sensitive	● Moderately robust	✓ Specialised fit	✓ Any	● Semi-parametric	⚠ Problematic

5 Evaluation

All of the code discussed in this section can be found at <https://github.com/Undeceiver/AbsolutelyMonotonicFitting/releases/tag/20260322>

We implemented and evaluated the following of the discussed methods:

- Constrained piecewise linear interpolation §4.2.
- Constrained Vandermonde fit §4.3.2.
- Convex splines §4.3.3.
- Greedy lowballing absolutely monotonic polynomial fit §4.3.4.

In order to compare their performance, we evaluate these on multiple source functions. For each source function, we produce a random uniform sample in $[0, 1]$, add random noise with either symmetric normal distribution or positive exponential distribution, and then run the fitting method on the resulting sample. We do multiple runs with different random samples for each source function, and measure the appropriate loss function paired to the type of noise introduced see §2, as well as visualizing the result. We measure the loss with respect to the noisy random sample used for the fit, and with respect to an equispaced noiseless sample, to check how well the characteristics of the source function are captured or how the method overfits to the noise.

We use a broad range of source functions to test the limits of each method:

- **Absolutely monotonic functions** - The core target of this paper.
 - **(Hyper)-exponentials** - Theoretically need infinite Taylor series but we should be able to approximate well with fewer parameters. Variety of more and less intense growth patterns.
 - * $f_1(x) = \frac{e^x - 1}{e - 1}$
 - * $f_2(x) = \frac{e^{e^x} - e}{e^e - e}$
 - * $f_3(x) = \frac{e^{e^{e^x}} - e^e}{e^{e^e} - e^e}$
 - **Absolutely monotonic polynomials** - These should be the best fits. We try a wide variety of degrees to test limits.
 - * $f_4(x) = x^2$
 - * $f_5(x) = x^5 + 2x^4 + x^3 + x$

$$\begin{aligned}
& * f_6(x) = 1.2x^4 + 0.37x^{50} + 0.1x^{71} \\
& * f_7(x) = x^{200}
\end{aligned}$$

- **Convex but not absolutely monotonic functions** - We evaluate these to see how well methods constrained to absolutely monotonic solutions can perform in them, where convex methods' additional flexibility should be beneficial.
 - **Convex polynomials** - Limited non-zero derivatives.
 - * $f_8(x) = 3x^2 - x^3$
 - * $f_9(x) = x^4 - 4x^3 + 6x^2$
 - **Trigonometric functions** - Infinitely differentiable and infinite non-zero derivatives, but not absolutely monotonic.
 - * $f_{10}(x) = \sin(\frac{\pi}{2}x - \frac{\pi}{2}) + 1$
 - **Piecewise polynomial** - Convex, growing, but second derivative is not continuous.
 - * $f_{11}(x) = 4x^2\{x \in [0, 0.5]\}, 4x - 1\{x \in [0.5, 1]\}$

6 Results and discussion

Figures 1 and 2 summarize the loss of the fitted function with respect to the source function with exponential and normal noise, respectively, and using asymmetric or symmetric loss, respectively, for each fitting approach. Emphasized are the best and worst values compared to other results for the same category. Figures 3 and 4 show this general comparison graphically on a logarithmic scale. Piecewise linear interpolation performs quite well in some cases but is only considered as a comparison element, since it fails to satisfy two of the most core requirements of our problem statement: smoothness and absolute monotonicity.

Apart from the basic properties of each approach as discussed in §4.4, key observations from the numerical simulations include:

- Constrained Vandermonde approximation is significantly worse than other approaches.
- Convex splines offer the best results when low degrees are sufficient, but fail dramatically with extreme growth patterns.

Figure 1: Loss with respect to source function (asymmetric exponential noise and loss function).

Source function	Piecewise linear §4.2	Constrained Vandermonde §4.3.2	Convex splines §4.3.3	GLAMPF §4.3.4
$f_1(x) = \frac{e^x - 1}{e - 1}$	1.02e-2	2.94e-2	<u>4.19e-3</u>	8.77e-3
$f_2(x) = \frac{e^{e^x} - e}{e^e - e}$	3.12e-3	<u>8.68e-2</u>	<u>1.63e-3</u>	2.52e-2
$f_3(x) = \frac{e^{e^e x} - e^e}{e^{e^e} - e^e}$	2.72e-2	<u>1.32e-1</u>	<u>1.04e-1</u>	<u>6.62e-3</u>
$f_4(x) = x^2$	8.77e-3	3.67e-2	<u>6.04e-3</u>	8.46e-3
$f_5(x) = x^5 + 2x^4 + x^3 + x$	2.14e-2	<u>1.62e-1</u>	<u>1.23e-2</u>	<u>1.22e-1</u>
$f_6(x) = 1.2x^4 + 0.37x^{50} + 0.1x^{71}$	<u>8.83e-3</u>	6.83e-2	4.82e-2	2.16e-2
$f_7(x) = x^{200}$	2.27e-2	<u>1.11e-1</u>	<u>1.11e-1</u>	<u>1.05e-2</u>
$f_8(x) = 3x^2 - x^3$	1.27e-2	<u>8.05e-2</u>	<u>6.44e-3</u>	<u>6.83e-2</u>
$f_9(x) = x^4 - 4x^3 + 6x^2$	1.12e-1	1.2e-1	<u>1.09e-2</u>	1.12e-1
$f_{10}(x) = \sin(\frac{\pi}{2}x - \frac{\pi}{2}) + 1$	3.76e-3	<u>8.24e-2</u>	<u>3.32e-3</u>	2.81e-2
$f_{11}(x) = 4x^2\{x \in [0, 0.5]\}, 4x - 1\{x \in [0.5, 1]\}$	8.45e-3	<u>1.36e-1</u>	<u>6.54e-3</u>	<u>1.4e-1</u>

- GLAMPF performs best, with a significant advantage, on functions with extreme growth patterns.
- GLAMPF is the most robust approach with asymmetric noise/loss on absolutely monotonic functions.
- GLAMPF is comparatively worse with symmetric noise/loss.
- GLAMPF is comparatively worse with non-absolutely monotonic functions.

The constrained Vandermonde approximation fit through non-linear least squares performs significantly worse than all other methods, especially with asymmetric noise/loss. A global optimization must simultaneously accommodate low and high degree structure with underestimation constraints, leading to systematic bias in one regime or the other. This can be observed in test cases by noticing that it either underestimates too much on lower values, which is better approximated with low degree polynomials (see figure 5), or it underestimates too much near 1, which is better approximated with high degree polynomials (see figure 6).

High degree dependence on functions with extreme growth patterns is particularly noticeable in the variation across methods. Convex splines perform best of all tested methods as long as high degrees are not required (see figures 7 and 8, for example), but lose significant representation capability

Figure 2: Loss with respect to source function (symmetric normal noise and loss function).

Source function	Piecewise linear §4.2	Constrained Vandermonde §4.3.2	Convex splines §4.3.3	GLAMPF §4.3.4
$f_1(x) = \frac{e^x - 1}{e - 1}$	3.03e-5	7.69e-5	<u>9.80e-6</u>	<u>1.38e-4</u>
$f_2(x) = \frac{e^{e^x} - e}{e^e - e}$	5.34e-5	1.70e-4	<u>1.29e-5</u>	<u>7.48e-4</u>
$f_3(x) = \frac{e^{e^e x} - e^e}{e^{e^e} - e^e}$	7.75e-4	<u>6.89e-2</u>	<u>9.81e-3</u>	<u>1.19e-4</u>
$f_4(x) = x^2$	1.35e-4	2.29e-4	<u>4.89e-5</u>	1.41e-4
$f_5(x) = x^5 + 2x^4 + x^3 + x$	<u>4.02e-3</u>	2.55e-2	4.35e-3	1.73e-2
$f_6(x) = 1.2x^4 + 0.37x^{50} + 0.1x^{71}$	2.45e-3	<u>1.63e-3</u>	6.26e-3	3.02e-3
$f_7(x) = x^{200}$	2.83e-2	<u>1.00e-1</u>	<u>6.88e-2</u>	<u>1.09e-3</u>
$f_8(x) = 3x^2 - x^3$	6.07e-5	<u>1.72e-3</u>	<u>3.50e-5</u>	<u>1.64e-3</u>
$f_9(x) = x^4 - 4x^3 + 6x^2$	8.95e-4	5.19e-3	<u>4.54e-4</u>	<u>1.02e-2</u>
$f_{10}(x) = \sin(\frac{\pi}{2}x - \frac{\pi}{2}) + 1$	8.28e-5	3.49e-4	<u>6.29e-5</u>	3.73e-4
$f_{11}(x) = 4x^2\{x \in [0, 0.5]\}, 4x - 1\{x \in [0.5, 1]\}$	<u>3.26e-4</u>	<u>7.11e-3</u>	3.64e-4	<u>7.99e-3</u>

Figure 3: Loss of various methods and source functions (asymmetric exponential noise and loss function).

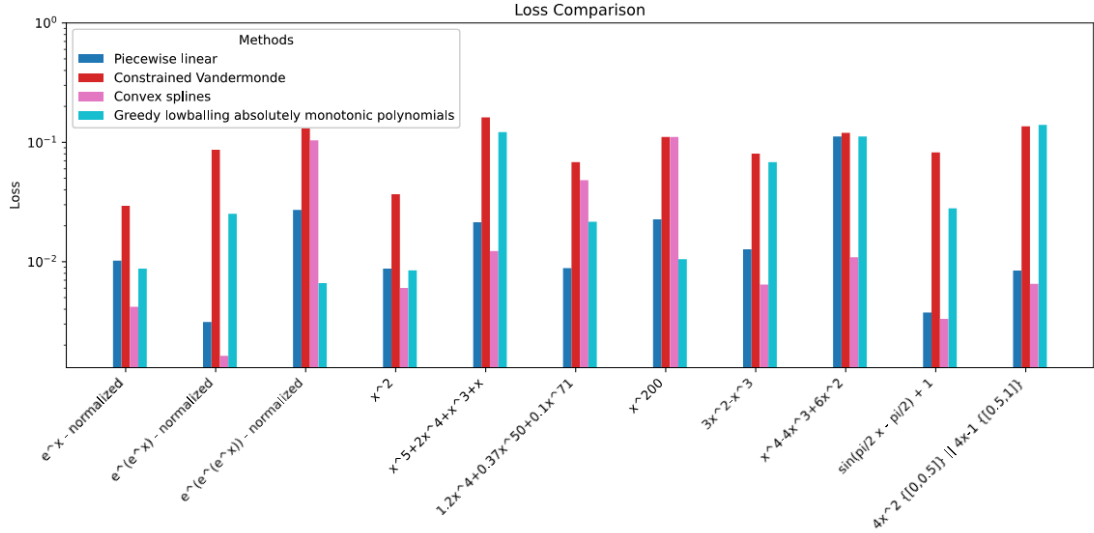


Figure 4: Loss of various methods and source functions (symmetric normal noise and loss function).

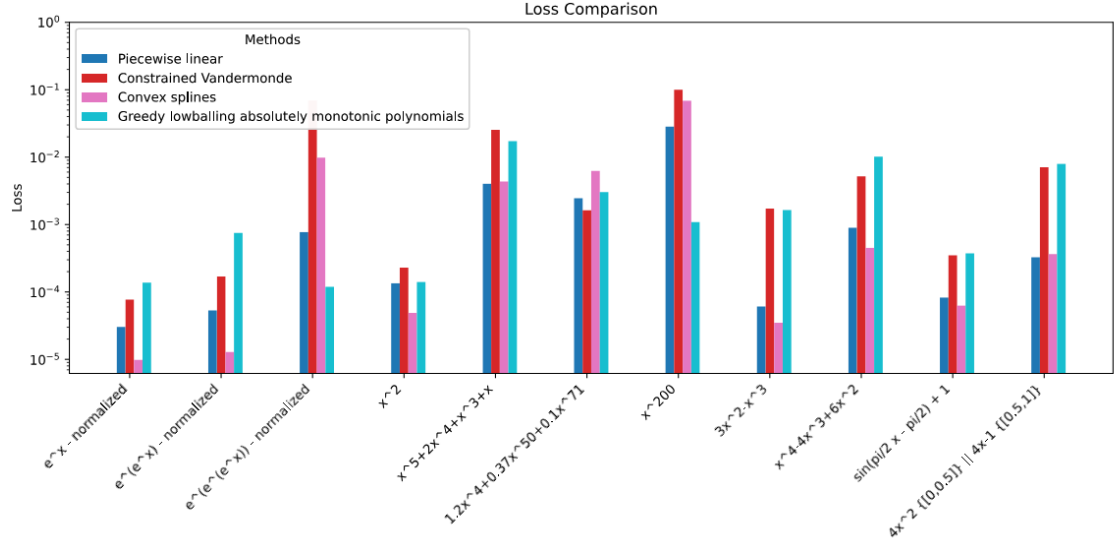


Figure 5: Vandermonde underestimates on lower values

Absolutely monotonic fit for x^2 (N=250, exponential noise mean=0.05)
 GLAMPF NMSE=7.45e-04, Loss=2.73e-02, GLAMPF Source NMSE=5.97e-05, Loss=8.46e-03
 Convex piecewise linear Fit NMSE=6.21e-04, Loss=2.74e-02, Convex piecewise linear Source NMSE=7.70e-05, Loss=8.77e-03
 Convex B-Spline Fit NMSE=6.27e-04, Loss=2.75e-02, Convex B-Spline Source NMSE=3.65e-05, Loss=6.04e-03
 Constrained Vandermonde Fit NMSE=3.20e-03, Loss=6.23e-02, Constrained Vandermonde Source NMSE=1.11e-03, Loss=3.67e-02

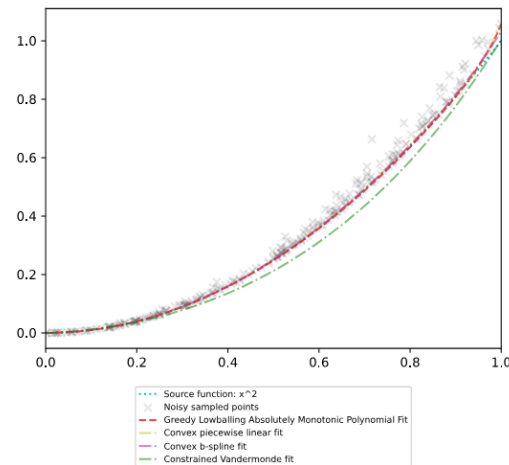
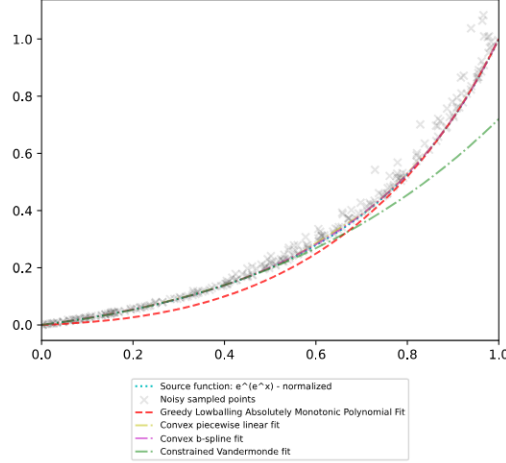


Figure 6: Vandermonde underestimates near 1

Absolutely monotonic fit for e^{e^x} - normalized (N=250, exponential noise mean=0.05)
 GLAMPF NMSE=2.12e-03, Loss=4.61e-02, GLAMPF Source NMSE=6.32e-04, Loss=2.52e-02
 Convex piecewise linear Fit NMSE=1.02e-03, Loss=3.51e-02, Convex piecewise linear Source NMSE=9.67e-06, Loss=3.12e-03
 Convex B-Spline Fit NMSE=1.06e-03, Loss=3.58e-02, Convex B-Spline Source NMSE=2.65e-06, Loss=1.63e-03
 Constrained Vandermonde Fit NMSE=1.13e-02, Loss=1.17e-01, Constrained Vandermonde Source NMSE=6.21e-03, Loss=8.68e-02



otherwise (figures 9, 10, 11, and 12). Conversely, GLAMPF performs particularly well compared with the other methods precisely in the cases where a high degree flexibility is required, while performing reasonably well in other cases.

Unsurprisingly, GLAMPF performs significantly worse in the non-absolutely monotonic cases, consistently struggling to align to the source curve in different regimes (high/low values), because of the absolutely monotonic shape constraint on the solution. See figures 13 and 14.

An issue that particularly affected GLAMPF in asymmetric noise/loss cases, is the overfitting to high noise values near 1 (tail). This is not an easy to solve issue, because this is fundamentally indistinguishable from high degree components in the underlying function's Taylor series. For intermediate values, noise can be avoided by bridging between lower and higher values in consistent ways, but when there are no higher values to bridge to, the noise can easily be interpreted as an additional high degree term. This is not merely an artifact of the algorithm, but an identifiability limitation inherent to the representation. This effect can be noticed for example in figure 13.

It is also worth pointing out that even in absolutely monotonic cases with difficult structure, GLAMPF clearly displays its non-optimality, giving no-

Figure 7: Convex splines perform particularly well when low degrees are sufficient

Absolutely monotonic fit for $x^5+2x^4+x^3+x$ (N=250, exponential noise mean=0.05)
 GLAMPF NMSE=7.09e-03, Loss=1.88e-01, GLAMPF Source NMSE=2.97e-03, Loss=1.22e-01
 Convex piecewise linear Fit NMSE=2.82e-03, Loss=1.31e-01, Convex piecewise linear Source NMSE=9.18e-05, Loss=2.14e-02
 Convex B-Spline Fit NMSE=3.04e-03, Loss=1.36e-01, Convex B-Spline Source NMSE=2.98e-05, Loss=1.23e-02
 Constrained Vandermonde Fit NMSE=1.10e-02, Loss=2.59e-01, Constrained Vandermonde Source NMSE=4.34e-03, Loss=1.62e-01

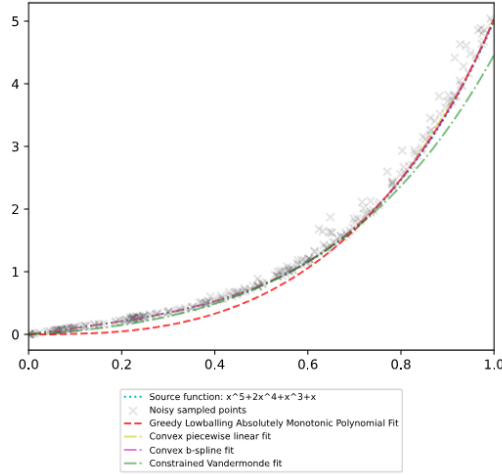


Figure 8: Convex splines perform particularly well when low degrees are sufficient

Absolutely monotonic fit for e^x - normalized (N=250, normal noise std=0.05)
 GLAMPF NMSE=5.80e-04, Loss=1.30e-03, GLAMPF Source NMSE=1.03e-04, Loss=1.38e-04
 Convex piecewise linear Fit NMSE=4.71e-04, Loss=1.28e-03, Convex piecewise linear Source NMSE=1.18e-05, Loss=3.03e-05
 Convex B-Spline Fit NMSE=4.83e-04, Loss=1.32e-03, Convex B-Spline Source NMSE=6.08e-06, Loss=9.80e-06
 Constrained Vandermonde Fit NMSE=5.08e-04, Loss=1.24e-03, Constrained Vandermonde Source NMSE=2.55e-05, Loss=7.69e-05

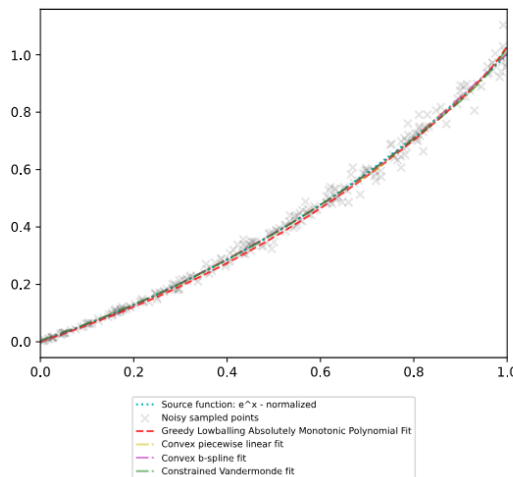


Figure 9: In extreme growth cases, GLAMPF significantly outperforms convex splines

Absolutely monotonic fit for $e^{(e^{(e^x)})}$ - normalized (N=250, exponential noise mean=0.05)
 GLAMPF NMSE=1.82e-05, Loss=4.26e-03, GLAMPF Source NMSE=3.83e-05, Loss=6.62e-03
 Convex piecewise linear Fit NMSE=8.67e-07, Loss=1.02e-03, Convex piecewise linear Source NMSE=6.17e-04, Loss=2.72e-02
 Convex B-Spline Fit NMSE=2.99e-03, Loss=6.03e-02, Convex B-Spline Source NMSE=8.82e-03, Loss=1.04e-01
 Constrained Vandermonde Fit NMSE=6.01e-03, Loss=8.54e-02, Constrained Vandermonde Source NMSE=1.42e-02, Loss=1.32e-01

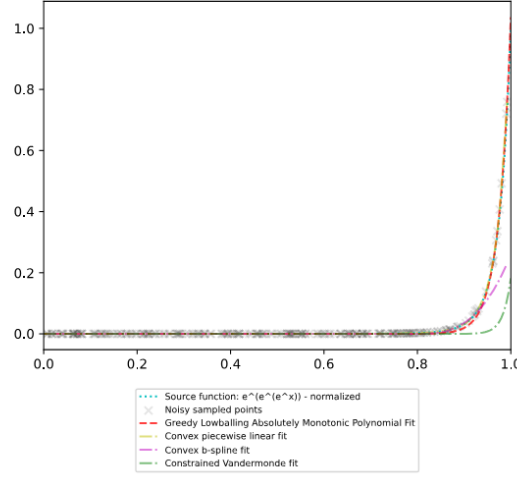


Figure 10: In extreme growth cases, GLAMPF significantly outperforms convex splines

Absolutely monotonic fit for $e^{(e^{(e^x)})}$ - normalized (N=250, normal noise std=0.05)
 GLAMPF NMSE=1.35e-04, Loss=6.66e-04, GLAMPF Source NMSE=3.92e-05, Loss=1.19e-04
 Convex piecewise linear Fit NMSE=1.02e-04, Loss=7.19e-04, Convex piecewise linear Source NMSE=8.25e-05, Loss=7.75e-04
 Convex B-Spline Fit NMSE=1.68e-03, Loss=7.36e-03, Convex B-Spline Source NMSE=2.09e-03, Loss=9.81e-03
 Constrained Vandermonde Fit NMSE=1.45e-02, Loss=7.16e-02, Constrained Vandermonde Source NMSE=1.07e-02, Loss=6.89e-02

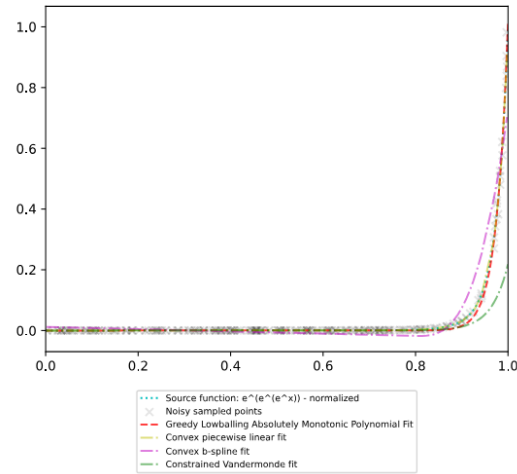


Figure 11: In extreme growth cases, GLAMPF significantly outperforms convex splines

Absolutely monotonic fit for x^{200} (N=250, exponential noise mean=0.05)
 GLAMPF NMSE=6.27e-06, Loss=2.50e-03, GLAMPF Source NMSE=9.13e-05, Loss=1.05e-02
 Convex piecewise linear Fit NMSE=4.14e-08, Loss=2.24e-04, Convex piecewise linear Source NMSE=4.25e-04, Loss=2.27e-02
 Convex B-Spline Fit NMSE=3.24e-03, Loss=6.29e-02, Convex B-Spline Source NMSE=1.02e-02, Loss=1.11e-01
 Constrained Vandermonde Fit NMSE=3.16e-03, Loss=6.20e-02, Constrained Vandermonde Source NMSE=1.00e-02, Loss=1.11e-01

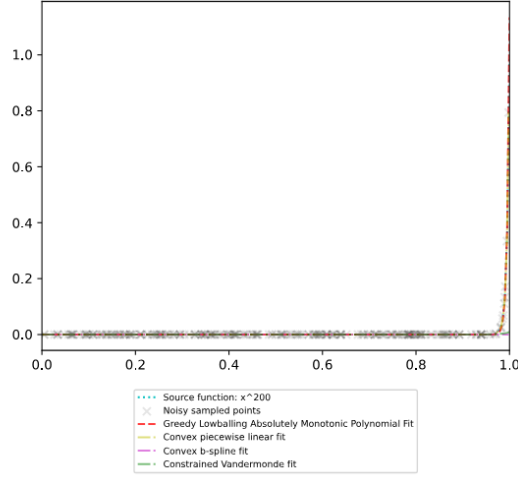


Figure 12: In extreme growth cases, GLAMPF significantly outperforms convex splines

Absolutely monotonic fit for x^{200} (N=250, normal noise std=0.05)
 GLAMPF NMSE=5.47e-06, Loss=5.81e-05, GLAMPF Source NMSE=8.02e-05, Loss=1.09e-03
 Convex piecewise linear Fit NMSE=4.11e-08, Loss=4.59e-07, Convex piecewise linear Source NMSE=2.84e-03, Loss=2.83e-02
 Convex B-Spline Fit NMSE=1.01e-03, Loss=7.09e-03, Convex B-Spline Source NMSE=7.33e-03, Loss=6.88e-02
 Constrained Vandermonde Fit NMSE=1.88e-03, Loss=1.82e-02, Constrained Vandermonde Source NMSE=1.02e-02, Loss=1.00e-01

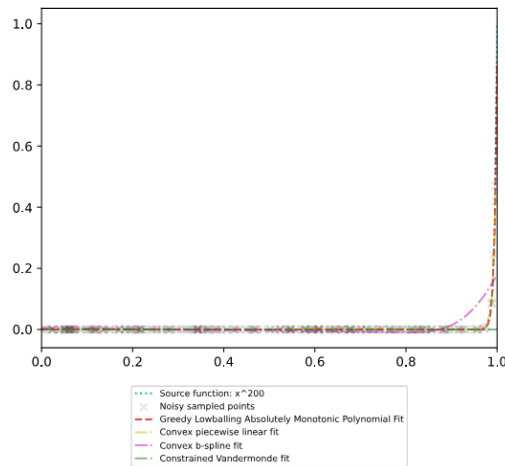


Figure 13: GLAMPF struggles to capture the entire regime when the source function is not absolutely monotonic

Absolutely monotonic fit for $3x^2 - x^3$ (N=250, exponential noise mean=0.05)
 GLAMPF NMSE=5.06e-03, Loss=1.01e-01, GLAMPF Source NMSE=2.02e-03, Loss=6.38e-02
 Convex piecewise linear Fit NMSE=1.14e-03, Loss=5.26e-02, Convex piecewise linear Source NMSE=8.03e-05, Loss=1.27e-02
 Convex B-Spline Fit NMSE=1.16e-03, Loss=5.30e-02, Convex B-Spline Source NMSE=2.06e-05, Loss=6.44e-03
 Constrained Vandermonde Fit NMSE=6.12e-03, Loss=1.22e-01, Constrained Vandermonde Source NMSE=2.67e-03, Loss=8.05e-02

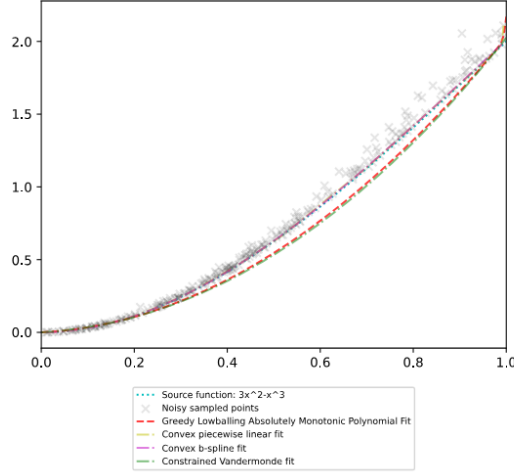


Figure 14: GLAMPF struggles to capture the entire regime when the source function is not absolutely monotonic

Absolutely monotonic fit for $4x^2 \{[0,0.5]\} \parallel 4x-1 \{[0.5,1]\}$ (N=250, normal noise std=0.05)
 GLAMPF NMSE=3.69e-03, Loss=2.03e-02, GLAMPF Source NMSE=2.17e-03, Loss=7.99e-03
 Convex piecewise linear Fit NMSE=1.58e-03, Loss=1.44e-02, Convex piecewise linear Source NMSE=5.80e-05, Loss=3.26e-04
 Convex B-Spline Fit NMSE=1.60e-03, Loss=1.45e-02, Convex B-Spline Source NMSE=5.53e-05, Loss=3.64e-04
 Constrained Vandermonde Fit NMSE=3.49e-03, Loss=1.83e-02, Constrained Vandermonde Source NMSE=1.82e-03, Loss=7.11e-03

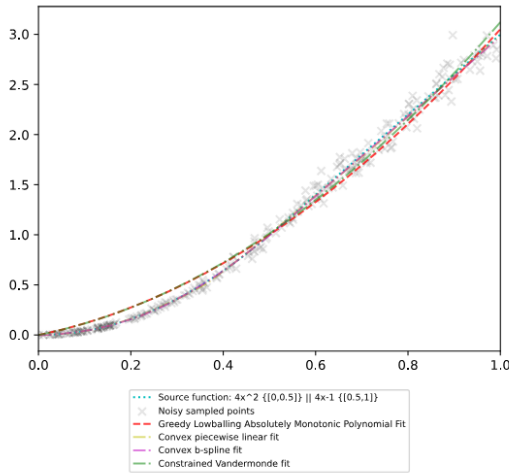
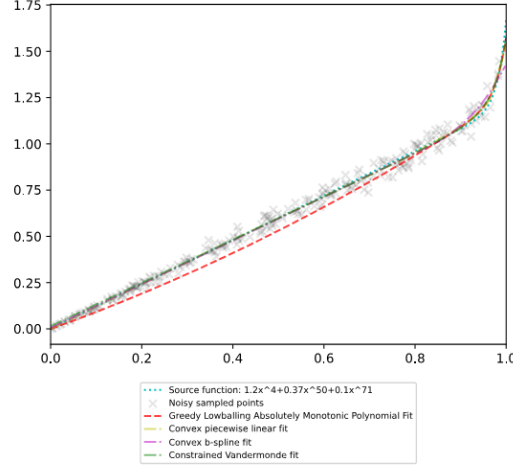


Figure 15: GLAMPF produces clearly non-optimal, but acceptable results

Absolutely monotonic fit for $1.2x^4 + 0.37x^{50} + 0.1x^{71}$ ($N=250$, normal noise std=0.05)
 GLAMPF NMSE=1.79e-03, Loss=4.39e-03, GLAMPF Source NMSE=1.44e-03, Loss=3.02e-03
 Convex piecewise linear Fit NMSE=4.99e-04, Loss=1.60e-03, Convex piecewise linear Source NMSE=1.88e-04, Loss=2.45e-03
 Convex B-Spline Fit NMSE=5.84e-04, Loss=1.94e-03, Convex B-Spline Source NMSE=5.18e-04, Loss=6.26e-03
 Constrained Vandermonde Fit NMSE=5.41e-04, Loss=1.62e-03, Constrained Vandermonde Source NMSE=1.61e-04, Loss=1.63e-03



ticeably worse solutions than other methods, but still avoiding catastrophic failure. See for example figure 15.

Considering that, out of the tested methods, constrained Vandermonde and GLAMPF are the only two that guarantee smoothness and absolute monotonicity, we consider these results provide good evidence of the appropriateness of GLAMPF when the shape constraints and/or robust behaviour for various kinds of functions, as it never performs catastrophically if the underlying source function is indeed absolutely monotonic. For less strict shape constraints and/or more limited growth functions, convex splines seems to be the best of the attempted approaches, though the non-attempted Bernstein polynomials method (§4.3.1) would likely produce similar results.

7 Conclusion

In §4, we discussed the reasons why polynomial methods are preferred to other methods for fitting absolutely monotonic functions in bounded intervals. Moreover, we justified why most of the existing approaches to this problem are not particularly well suited. We evaluated a few of the approaches that were potentially promising. Convex splines perform extremely well as long as the underlying function does not have fast growth patterns that require high degree polynomials to represent. Our newly introduced method of

greedy lowballing absolutely monotonic polynomial fitting shows significant flexibility to all types of absolutely monotonic functions and symmetric or asymmetric loss functions; as well as acceptable results on other convex but not absolutely monotonic functions. It is particularly effective at detecting fast growth patterns towards the higher end of the interval. However, it does not produce optimal results in most cases and frequently shows noticeable error across the different regimes due to the rigidity of the solutions.

The significant differences in results of the various methods linked to the growth patterns suggest ideas for future work that combine the strengths of multiple approaches. For example, a hybrid algorithm that attempts multiple methods, including some not implemented here, and selects the one that performed best for the particular data presented, could potentially exploit the strengths of the various approaches. Similarly, the greedy approach could be extended to consider exponential or hyperexponential terms, under the change of variables discussed in §3 and/or in combination with polynomial terms in the same sum. Dynamically adjusting the degree of splines or Bernstein polynomials, or dynamically adjusting growth constraints in some of these methods, could lead to a broader and more effective search process that finds a more suitable representation of the data in a wider variety of cases.

The issues with the approaches described here can be summarized in two families: unsuitable constraints and assumptions for the specific underlying function, and non-optimal methods that do not produce the best results possible. This contrast suggests that there may be a trade-off between expressivity for high-growth behavior and stability / optimality under constraints. The combination of both or the dynamic adjustment of constraints and assumptions depending on results could potentially find a better balance in more diverse cases.

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¹in particular, OpenAI ChatGPT (models available through the ChatGPT free public service at the time of use)

References

- [Aguilera and Morin, 2008] Aguilera, N. E. and Morin, P. (2008). Approximating optimization problems over convex functions. *Numerische Mathematik*, 111:1–34.
- [Bernstein, 1912] Bernstein, S. (1912). Démonstration du théorème de weierstrass fondée sur le calcul des probabilités. 13(1):1–2.
- [Bernstein, 1929] Bernstein, S. (1929). Sur les fonctions absolument monotones. *Acta Mathematica*, 52(1):1–66.
- [Björck and Pereyra, 1970] Björck, A. and Pereyra, V. (1970). Solution of vandermonde systems of equations. *Mathematics of computation*, 24(112):893–903.
- [Boyd and Vandenberghe, 2004] Boyd, S. and Vandenberghe, L. (2004). *Convex optimization*. Cambridge university press.
- [Chunaev and Safiullin, 2020] Chunaev, P. and Safiullin, I. (2020). Data fitting by exponential sums with equal weights. In Krzhizhanovskaya, V. V., Závodszy, G., Lees, M. H., Dongarra, J. J., Soot, P. M. A., Brissos, S., and Teixeira, J., editors, *Computational Science – ICCS 2020*, pages 364–371, Cham. Springer International Publishing.
- [Evans et al., 1980] Evans, J. W., Gragg, W. B., and LeVeque, R. J. (1980). On least squares exponential sum approximation with positive coefficients. *Mathematics of Computation*, 34(149):203–211.
- [Ghosal et al., 2023] Ghosal, R., Ghosh, S., Urbanek, J., Schrack, J. A., and Zipunnikov, V. (2023). Shape-constrained estimation in functional regression with bernstein polynomials. *Computational Statistics & Data Analysis*, 178:107614.
- [Groeneboom et al., 2001] Groeneboom, P., Jongbloed, G., and Wellner, J. A. (2001). Estimation of a convex function: characterizations and asymptotic theory. *The Annals of Statistics*, 29(6):1653–1698.
- [Hannah and Dunson, 2013] Hannah, L. A. and Dunson, D. B. (2013). Multivariate convex regression with adaptive partitioning. *J. Mach. Learn. Res.*, 14(1):3261–3294.
- [He and Shi, 1998] He, X. and Shi, P. (1998). Monotone b-spline smoothing. *Journal of the American Statistical Association*, 93(442):643–650.

- [Holmström and Petersson, 2002] Holmström, K. and Petersson, J. (2002). A review of the parameter estimation problem of fitting positive exponential sums to empirical data. *Appl. Math. Comput.*, 126(1):31–61.
- [Kammler, 1976] Kammler, D. W. (1976). Chebyshev approximation of completely monotonic functions by sums of exponentials. *SIAM Journal on Numerical Analysis*, 13(5):761–774.
- [Keller and Plonka, 2021] Keller, I. and Plonka, G. (2021). Modifications of prony’s method for the recovery and sparse approximation with generalized exponential sums. In Fasshauer, G. E., Neamtu, M., and Schumaker, L. L., editors, *Approximation Theory XVI*, pages 123–152, Cham. Springer International Publishing.
- [Kopotun et al., 2011] Kopotun, K., Leviatan, D., Prymak, A., and Shevchuk, I. (2011). Uniform and pointwise shape preserving approximation by algebraic polynomials. *arXiv preprint arXiv:1109.0968*.
- [Koumandos, 2014] Koumandos, S. (2014). On completely monotonic and related functions. In Rassias, T. M. and Pardalos, P. M., editors, *Mathematics Without Boundaries: Surveys in Pure Mathematics*, pages 285–321. Springer New York, New York, NY.
- [Lachand-Robert and Oudet, 2005] Lachand-Robert, T. and Oudet, E. (2005). Minimizing within convex bodies using a convex hull method. *SIAM Journal on Optimization*, 16(2):368–379.
- [Lebair and Shen, 2015] Lebair, T. M. and Shen, J. (2015). Uniform lipschitz property of nonnegative derivative constrained b-splines and applications to shape constrained estimation. *arXiv preprint arXiv:1510.05314*.
- [Lu, 2015] Lu, M. (2015). Spline estimation of generalised monotonic regression. *Journal of Nonparametric Statistics*, 27(1):19–39.
- [Nagahara and Martin, 2013] Nagahara, M. and Martin, C. F. (2013). Monotone smoothing splines using general linear systems. *Asian Journal of Control*, 15(2):461–468.
- [Smith et al., 1976] Smith, M. R., Cohn-Sfetcu, S., and Buckmaster, H. A. (1976). Decomposition of multicomponent exponential decays by spectral analytic techniques. *Technometrics*, 18(4):467–482.
- [Wang and Ghosh, 2012] Wang, J. and Ghosh, S. (2012). Shape restricted nonparametric regression with bernstein polynomials. *Computational Statistics & Data Analysis*, 56(9):2729–2741.

[Zhang, 2004] Zhang, J.-T. (2004). A simple and efficient monotone smoother using smoothing splines. *Journal of Nonparametric Statistics*, 16(5):779–796.

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